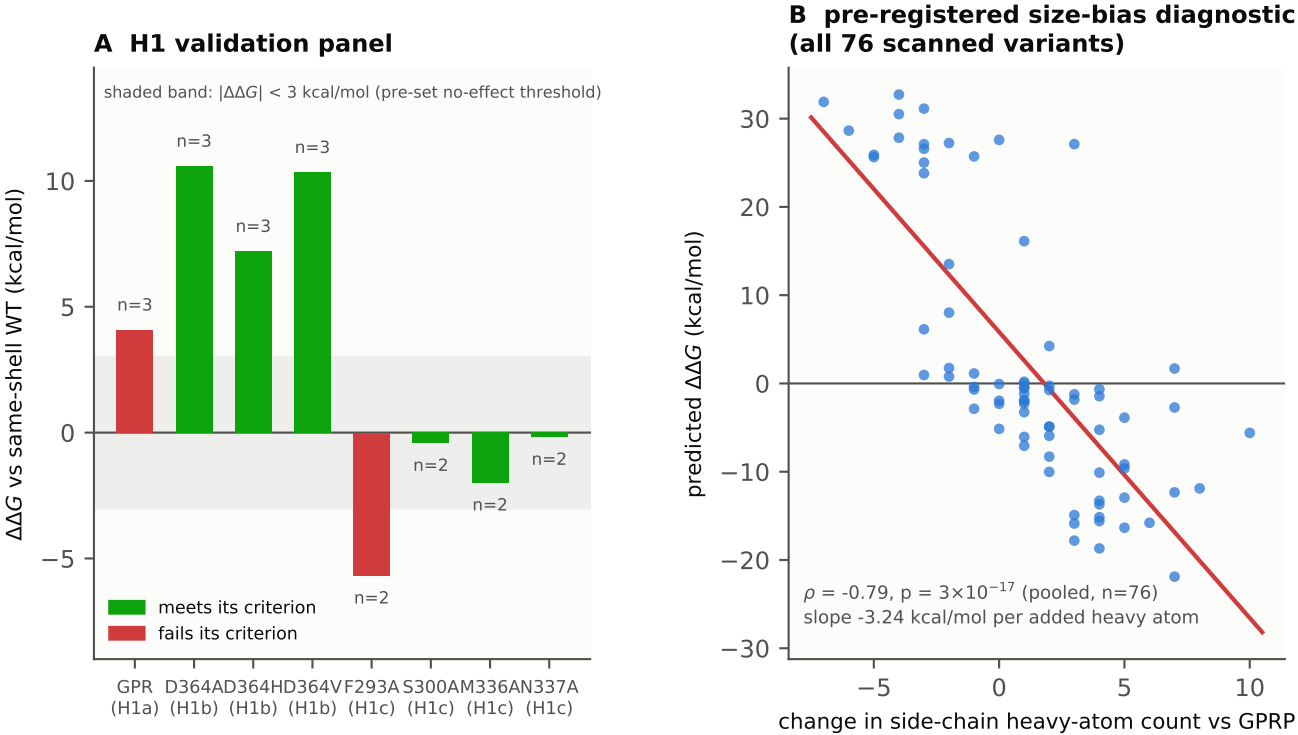
A H1 validation panel
shaded band: |ΔΔG| < 3 kcal/mol (pre-set no-effect threshold)
ΔΔG vs same-shell WT (kcal/mol)
10
5
0
−5
n=3
n=3
n=3
n=3
n=2
n=2
n=2
n=2
meets its criterion
fails its criterion
GPR (H1a)
D364A (H1b)
D364H (H1b)
D364V (H1b)
F293A (H1c)
S300A (H1c)
M336A (H1c)
N337A (H1c)
B pre-registered size-bias diagnostic
(all 76 scanned variants)
predicted ΔΔG (kcal/mol)
30
20
10
0
−10
−20
−30
−5
0
5
10
ρ = -0.79, p = 3×10−17 (pooled, n=76)
slope -3.24 kcal/mol per added heavy atom
change in side-chain heavy-atom count vs GPRP